which is an integer. It follows that $\frac{510}{n+300}$ is an integer which is obviously positive and is less than $\frac{510}{300}$ i.e

it is less than 2. The only possibility is that $\frac{510}{n+300} = 1 \Longrightarrow n + 300 = 510$. Therefore $n = 210$. Hence there are 210 red boxes in the storage.

Qn15. Let $\overline{PB} = x, \overline{PQ} = y, \overline{BQ} = \overline{QC} = 2$. Since $< PQR = 60°$, it implies $< PQB + < RQC = 120°$ so that $< ACB = 60°$. Further more, $< RQC + < CRQ = 120°$ so that $< PQB =< CRQ$.

Let this angle be α and using the sine rule on $\triangle PBQ$, we have

$$\frac{PB}{\sin\alpha} = \frac{PQ}{\sin 90} \Longrightarrow \frac{x}{\sin\alpha} = y \Longrightarrow x = y\sin\alpha.$$

On $\triangle QCR$, we have

$$\frac{QC}{\sin\alpha} = \frac{RQ}{\sin 60} \Longrightarrow \frac{2}{\sin\alpha} \times \frac{y}{\frac{\sqrt{3}}{2}} \Longrightarrow y = \frac{\sqrt{3}}{\sin\alpha}.$$

Therefore

$$x = y\sin\alpha = \frac{\sqrt{3}}{\sin\alpha} \times \sin\alpha = \sqrt{3}.$$

We note by Pythagoras Theorem that

$$(\sqrt{3})^2 + 2^2 = (PQ)^2 \Longrightarrow PQ = \sqrt{3+4} = \sqrt{7} = PR.$$

Therefore $\overline{PR} = \sqrt{7}$.

NMC 2016 A-LEVEL PAPER 2 QUESTIONS

SECTION A

Qn1. Let a be a number such that $\log_a(10) + \log_a(10^2) + \log_a(10^3) + \ldots + \log_a(10^{10}) = 110$. What is the value of a?

Qn2. What is the smallest positive integer x for which $\frac{1}{32} = \frac{x}{10^y}$ for some positive integer y?

Qn3. A sequence is defined by $a_1 = 1$ and for $n \geq 1$ by;

$$a_{n+1} = \begin{cases} 0 & \text{if } a_n = 0 \text{ and } n \textit{ is} \text{ odd} \\ 2 & \text{if } a_n = 0 \text{ and } n \textit{ is} \text{ even} \\ 1 & \text{if } a_n = 1 \text{ and } n \textit{ is} \text{ odd} \\ 0 & \text{if } a_n = 1 \text{ and } n \textit{ is} \text{ even} \\ 1 & \text{if } a_n = 2 \end{cases}$$

How many of the numbers $a_1, a_2, a_3, \ldots, a_{2016}$ are equal to 2?

Qn4. How many members of the set $\{\frac{1}{7}, \frac{2}{7}, \frac{3}{7}, \ldots, \frac{99}{7}\}$ are integer multiples of $\frac{2}{5}$?

Qn5. Find all ordered pairs (x, y) of real numbers satisfying the following equations:

$$x^{2016} + 2y^{2016} = x^{2015} + 2y^{2015} = x^{2014} + 2y^{2014}.$$

Qn6. Consider the equation $e^y + e^{-y} = 5$. The largest solution of the equation can be written in the form $y = \ln\left(\frac{A+\sqrt{B}}{C}\right)$. Find the value of $A + B + C$.

Qn7. Circles of radius 10 and 17 intersect at two points. The segment connecting these points of intersection has length 16. List all possible values for the distance between the centers of the circles?

Qn8. How many of the 2-digit numbers from 10 to 99 have the property that both digits are perfect squares? For example, 10 is the smallest such a number and 99 is the largest.

Qn9. Let $x = \sqrt{16 + \sqrt{16 + \sqrt{\cdots}}}$. What is the value of x?

Qn10. In a group of dogs and people, the number of legs was 28 more than twice the number of heads. How many dogs were there? [Assume none of the people or dogs is missing a leg.]

SECTION B

Qn11. On a particular street in Muyenga, there are exactly 14 houses, each numbered with an integer between 500 and 599, inclusive. The 14 house numbers form an arithmetic sequence in which 7 terms are even and 7 terms are odd. One of the houses is numbered 555 and none of the remaining 13 numbers has two equal digits. What is the smallest of the 14 house numbers?

Qn12. (a) Find all real numbers x for which

$$\frac{8^x + 27^x}{12^x + 18^x} = \frac{7}{6}$$

(b) The function f, defined by

$$f(x) = \frac{ax + b}{cx + d}, \quad x \neq -\frac{d}{c}$$

where a, b, c, and d are nonzero real numbers, has the properties

$$f(19) = 19, \ f(97) = 97, \text{ and } f(f(x)) = x,$$

for all values of x. Find the range of f.

Qn13. A strip of rubber is 80 cm long initially, and after every minute it is instantaneously and uniformly stretched by 40 cm. An ant moves at a rate of 42 cm per minute, starting at one end. Each time the strip is stretched, the ant is moved to a position on the modified strip proportional to its position on the strip before the stretching took place. How many minutes does it take the ant to cross the strip?

Qn14. Starting with an equilateral triangle, you inscribe a circle in the triangle, and then inscribe an equilateral triangle inside the circle. You then repeat this process four more times, each time inscribing a circle and then an equilateral triangle in the smallest triangle constructed up to that point, so that you end up drawing five triangles in addition to the one you started with. What is the ratio of of the area of the largest triangle to the area of the smallest triangles?

Qn15. Alistair, Conrad, Emma, and Salma compete in a three-sport race. They each swim 2 km, then bike 40 km, and finally run 10 km. Also, they each switch instantly from swimming to biking and from biking to running.

(a) Emma has completed $\frac{1}{13}$ of the total distance of the race. How many kilometers has she travelled?

(b) Conrad began the race at 8 : 00 a.m. and completed the swimming portion in 30 minutes. Conrad biked 12 times as fast as he swam, and ran 3 times as fast as he swam. At what time did he finish the race?

(c) Alistair and Salma also began the race at 8 : 00 a.m. Alistair finished the swimming portion in 36 minutes, and then biked at 28 km/h. Salma finished the swimming portion in 30 minutes, and then biked at 24 km/h. Alistair passed Salma during the bike portion. At what time did Alistair pass Salma?

NMC 2016 A-LEVEL PAPER 2 SOLUTIONS

SECTION A

Qn1.

$$\begin{aligned} \log_a(10) + \log_a(10^2) + \log_a(10^3) + \cdots + \log_a(10^{10}) &= 110 \\ \log_a\left(10 \times 10^2 \times 10^3 \times \cdots \times 10^{10}\right) &= 110 \\ \log_a\left(10^{1+2+3+\cdots+10}\right) &= 110 \\ \log_a\left(10^{55}\right) &= 110 \\ 55 \log_a 10 &= 110 \\ \log_a 10 &= 2 \end{aligned}$$

$$a^2 = 10 \quad \Rightarrow \quad a = \sqrt{10}$$

Qn2. Since $10^y \neq 0$, the equation $\frac{1}{32} = \frac{x}{10^y}$ is equivalent to $10^y = 32x$.

So the given question is equivalent to asking for the smallest positive integer x for which $32x$ equals a positive integer power of 10.

Now $32 = 2^5$ and so $32x = 2^5 x$.

For $32x$ to equal a power of 10, each factor of 2 must be matched with a factor of 5.

Therefore, x must be divisible by 5^5 (that is, x must include at least 5 powers of 5), and so

$$x \geq 5^5 = 3125$$

But $32(5^5) = 2^5 5^5 = 10^5$, and so if $x = 5^5 = 3125$, then $32x$ is indeed a power of 10, namely 10^5. This tells us that the smallest positive integer x for which $\frac{1}{32} = \frac{x}{10^y}$ for some positive integer y is $x = 5^5 = 3125$.

Qn3. Starting with $a_1 = 1$, the sequence goes as

$$1, 1, 0, 0, 2, 1, 0, 0, 2, \ldots$$

and keeps repeating with period 4, that is $a_n = 2$ if and only if $n = 5, 9, \cdots$

1 greater than a positive multiple of 4 thus; a_{2016} will be the 503rd 2 in $a_1, a_2, a_3, \ldots, a_{2016}$. So 503.

Qn4. Let $\frac{a}{7} = k \cdot \frac{2}{5}$ where k is an integer, then $5a = 14k \quad \Rightarrow \quad a$ must be a multiple of 14. The only multiples of 14 below 99 are

$$14, 28, 42, 46, 70, 84, 98$$

Thus there are 7 such numbers less than 100.

Qn5. Solving the three equations

$$\begin{aligned} x^{2016} + 2y^{2016} &= x^{2015} + 2y^{2015} \\ x^{2016} + 2y^{2016} &= x^{2014} + 2y^{2014} \\ x^{2015} + 2y^{2015} &= x^{2014} + 2y^{2014} \end{aligned}$$

Gives the solutions $(x, y) = (0, 0),\ (0, 1),\ (1, 0),\ (1, 1)$

Qn6. Solving the equation

$$\begin{aligned} e^y + e^{-y} &= 5 \\ e^y + \frac{1}{e^y} &= 5 \\ (e^y)^2 + 1 &= 5e^y \\ (e^y)^2 - 5e^y + 1 &= 0 \end{aligned}$$

A quadratic equation in e^y, whose solution is given by

$$e^y = \frac{5 \pm \sqrt{25 - 4(1)(1)}}{2} = \frac{5 \pm \sqrt{21}}{2}$$

With a maximum value of

$$e^y = \frac{5 + \sqrt{21}}{2} \quad \Rightarrow \quad y = \ln\left(\frac{5 \pm \sqrt{21}}{2}\right)$$

The largest solution of the equation can be written in the form

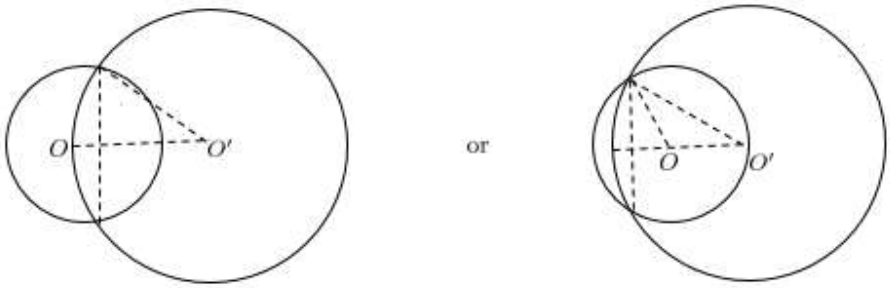
$$y = \ln\left(\frac{A + \sqrt{B}}{C}\right).$$

To have the value of $A + B + C = 5 + 21 + 2 = 28$.

Qn7. There are two options of drawing the two circles The line connecting the centers O and O' is the perpendicular bisection of this chord, intersecting it at point P.

If O and O' are centers of the circles, then

$$OO' = PO' \pm OP$$



and OP and PO' are bases of right triangles with height 8 and hypotenuse 10 and 17 respectively.

By Pythagoras theorem, OP and PO' are 6 and 15 respectively. Thus

$$OO' = PO' \pm OP = 15 \pm 6 = 9 \text{ units} \ , \ 2 \text{ units}$$

Qn8. The only perfect square single digit numbers are $0, 1, 4, 9$

A 2-digit number can be formed from the set $\{0, 1, 4, 9\}$ only starting with $1, 4, 9$. That is three possible choices to start a 2-digit perfect square.

The next digit could take on any of the four digits since repetition is allowed. That is

$$3 \times 4 = 12$$

Qn9. The value of x can be found by solving the algebraic equation $x = \sqrt{16 + x}$. This gives $x^2 - x - 16 = 0$, and the positive root given by the quadratic formula is

$$x = \frac{1}{2} + \frac{\sqrt{65}}{2}$$

The other root is extraneous.

Remark: For this problem to have a real solution, it is essential for the associated sequence

$$a_{n+1} = \sqrt{16 + a_n}$$

to converge. To show this sequence does converge we use the fact that every nonempty set of real numbers that is bounded above has a least upper bound. That the sequence $\{a_n\}$ is increasing and bounded above (by 5) follows by an easy induction proof.

Qn10. If there were only people, there would be exactly twice the number of legs as heads. Each dog contributes two extra legs over and above this number. So 28 extra legs means 14 dogs.

SECTION B

Qn11. Since this arithmetic sequence of integers includes both even and odd integers, then the common difference in the sequence must be odd. (If it were even, then all of the terms in the sequence would be even or all would be odd.)

Suppose that this common difference is d. Note that 555 is one of the terms in the sequence.

If $d = 1$, then either or both of 554 and 556 would be in the sequence. Since no term other than 555 can contain

repeated digits, then $d \neq 1$.

If $d = 3$, then either or both of 552 and 558 would be in the sequence. Since no term other than 555 can contain repeated digits, then $d \neq 3$.

If d = 5, then 550 cannot be in the sequence, so 555 would have to be the smallest term in the sequence. This would mean that 560 and 565 were both in the sequence.

Since no term other than 555 can contain repeated digits, then $d \neq 5$.

We can also rule out the possibility that $d \geq 9$: [Since there are 14 terms in the sequence and the common difference is d, then the largest term is 13d larger than the smallest term. Since all terms are between 500 and 599, inclusive, then the maximum possible difference between two terms is 99. This means that $13d \leq 99$. Since d is an odd integer, then cannot be 9 or greater.]

Since d is odd, d cannot be 1, 3 or 5, and $d \leq 7$, then it must be the case that $d = 7$. Can we actually construct a sequence with the desired properties? Starting with 555 and adding and subtracting 7s, we get the sequence

$$506,\ 513,\ 520,\ 527,\ 534,\ 541,\ 548,\ 555,\ 562,\ 569,\ 576,\ 583,\ 590,\ 597$$

This sequence has 14 terms in the desired range, including 7 even and 7 odd terms, and no term other than 555 has repeated digits. (Note that this sequence cannot be extended in either direction and stay in the desired range.) Therefore, the smallest of the 14 house numbers must be 506.

Qn12. (a) By setting $2^x = a$ and $3^x = b$, the equation becomes - *common parameters*

$$\begin{aligned} \frac{a^3 + b^3}{a^2b + b^2a} &= \frac{7}{6} \\ \\ \frac{a^2 - ab + b^2}{ab} &= \frac{7}{6} \\ \\ 6a^2 - 13ab + 6b^2 &= 0 \\ (2a - 3b)(3a - 2b) &= 0 \end{aligned}$$

Therefore $2^{x+1} = 3^{x+1}$ or $2^{x-1} = 3^{x-1}$, which implies that $x = -1$ and $x = 1$

It is easy to check that both $x = -1$ and $x = 1$ satisfy the equation

(b) For all x, - *we start with a more general option*

$$\begin{aligned} \frac{a\left(\frac{ax+b}{cx+d}\right) + b}{c\left(\frac{ax+b}{cx+d}\right) + d} &= x \\ \\ \frac{(a^2 + bc)x + b(a + d)}{c(a + d)x + bc + d^2} &= x \\ \\ c(a + d)x^2 + (d^2 - a^2)x - b(a + d) &= 0, \end{aligned}$$

Wrong: equating coefficients of x,
Wrong: solving a quadratic in x, no need for x

But

$$\begin{aligned} c(a+d)x^2 + (d^2-a^2)x - b(a+d) &= 0 \\ (a+d)\left[cx^2 + (d-a)x - b\right] &= 0 \end{aligned}$$

which implies that $c(a+d)=0$. Since $c \neq 0$, we must have $a=-d$.

The conditions $f(19)=19$ and $f(97)=97$ lead to the equations

$$19^2c = 2\cdot 19a + b \text{ and } 97^2c = 2\cdot 97a + b$$

Hence $(97^2-19^2)c = 2(97-19)a$
It follows that $a=58c$, which in turn leads to $b=-1843c$. Therefore

$$f(x) = \frac{58x-1843}{x-58} = 58 + \frac{1521}{x-58}$$

which never has the value 58.

Thus the range of f is $\mathbb{R} - \{58\}$

Qn13. After one minute, it will be $\frac{42}{80}$ of the length on the rubber strip.

After stretching, it will be at $\frac{63}{120}$ because he stretch is by $40cm$ from $80cm$ which imply an increase by $\frac{1}{2}$ of the length. So the proportional stretch is $\frac{1}{2}\times 42 = 21$, so $42+21=63$.

After the second minute, it will be at $\frac{105}{120}$, that is 42×63 along the rubber.

After the second stretching, it will be at $\frac{140}{160}$. Since the stretching is from 120 to 160 which is $\frac{1}{3}$ of 120, the proportional stretch is $\frac{1}{3}\times 105 = 35$, added to 105 to get 140.

The remaining $\frac{20}{160}$ is covered in $\frac{20}{42}$ minutes, that is a total time is $2\frac{20}{40}$ minutes.

Qn14. Each time you draw a new triangle, it has its vertices at the midpoints of the three sides of the old triangle. The new triangle is similar to the old triangle, and indeed this process divides the old triangle into four smaller triangles which are congruent and which each have area $\frac{1}{4}$ of the old triangle. The answer is therefore $4^5 = 1024$.

Qn15. (a) When Emma has completed $\frac{1}{13}$ of the total distance, she has travelled $\frac{1}{13}\times 52 = 4$ km

(b) Since Conrad completed the 2 km swim in 30 minutes (which is half an hour), then his speed was $2 \div \frac{1}{2} = 4$ km/h Since Conrad biked 12 times as fast as he swam, then he biked at $12\times 4 = 48$ km/h.
Since Conrad biked 40 km, then the bike portion took him $\frac{40}{48} = \frac{5}{6}$ *hours*.
Since 1 hour equals 60 minutes, then the bike portion took him $\frac{5}{6}\times 60 = 50$ minutes.
Since Conrad ran 3 times as fast as he swam, then he ran at $3\times 4 = 12$ km/h.
Since Conrad ran 10 km, then the running portion took him $\frac{10}{12} = \frac{5}{6}$ hours.
This is again 50 minutes.

Therefore, the race took him $30+50+50 = 130$ minutes, or 2 hours and 10 minutes. Since Conrad began the race at 8 : 00 a.m., then he completed the race at 10 : 10 a.m.

(c) Suppose that Alistair passed Salma after t minutes of the race. Since Alistair swam for 36 minutes, then he had biked for $t-36$ minutes (or $\frac{t-36}{60}$ hours)when they passed.

Since Salma swam for 30 minutes, then she had biked for $t-30$ minutes (or $\frac{t-30}{60}$ hours) when they passed.

When Alistair and Salma passed, they had travelled the same total distance. At this time, Alistair had swum 2 km. Since he bikes at 28 km/h, he had biked $28\times\frac{t-36}{60}$ km.

Similarly, Salma had swum 2 km. Since she bikes at 24 km/h, she had biked $24\times\frac{t-30}{60}$ km.

Since their total distances are the same, then

$$\begin{aligned} 2 + 28 \times \frac{t-36}{60} &= 2 + 24 \times \frac{t-30}{60} \\ 28 \times \frac{t-36}{60} &= 24 \times \frac{t-30}{60} \\ 28(t-36) &= 24(t-30) \\ 7(t-36) &= 6(t-30) \\ 7t - 252 &= 6t - 180 \\ t &= 72 \end{aligned}$$

Therefore, Alistair passed Salma 72 minutes into the race. Since the race began at 8 : 00 a.m., then he passed her at 9 : 12 a.m.

Alternatively, we could notice that Salma started the bike portion of the race 6 minutes before Alistair. Since 6 minutes is $\frac{1}{10}$ of an hour, she biked a distance of $\frac{1}{10} \times 24 = 2.4$ km before Alistair started riding.

In other words, she has a 2.4 km head start on Alistair. Since Alistair bikes 4 km/h faster than Salma, it will take $\frac{2.4}{4} = 0.6$ hours for him to make up this difference.

Since 0.6 hours equals $0.6 \times 60 = 36$ minutes and he swam for 36 minutes before starting the bicycle portion, he will catch up a total of $36 + 36 = 72$ minutes into the race.